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SECTION: G

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Topic: Alumni Mentorship & Mock Interview Platform

ARCHITECTURE SELECTION & WRITTEN JUSTIFICATION

Architecture Selection

We chose Layered Architecture for the Alumni Mentorship & Mock Interview Platform.

Architectural Choice

The platform is organized into Presentation, Application/Business, and Data/Integration responsibilities. This provides clear separation of concerns while supporting the tightly connected workflow of mentor recommendation, availability checking, booking, mock-interview scorecards, and session history.

Two Reasons

1. The platform contains a sequential business workflow in which recommendations, availability, booking, scorecards, and session history share consistent user and session information. A layered design keeps these business rules centralized and easier to maintain.
2. The system is primarily a single integrated platform with external calendar/meeting integration. Layered architecture is simpler to develop, test, deploy, and understand than a distributed microservices design, while still allowing the external service to be isolated behind an integration interface.

Security Advantage

Authentication and authorization can be enforced at the presentation/application boundary, while the data layer can restrict access to profile data and scorecards. This supports the requirement that deleted data is inaccessible to unauthorized users and that scorecards are visible only to the corresponding student.

Performance Benefit

The application/business layer can optimize mentor recommendation and availability queries using indexed data and caching while keeping the data layer efficient. This supports the requirement that at least 95% of recommendation and availability requests complete within 3 seconds under the defined normal operating load.
Architecture comparison: Layered was selected over Microservices and Client-Server because it offers the best balance of maintainability, consistency, security, and implementation simplicity for the stated platform. Microservices provide independent scaling and fault isolation but introduce operational complexity, network latency, and data-consistency challenges; Client-Server offers centralized control but has a scalability bottleneck and single-point-of-failure concerns.

Component note

The handout's Order Manager and Payment Service are written for its example coffee-kiosk scenario. They are not supported by the assigned Alumni Mentorship requirements, so they are replaced by Booking & Session Management and Mentor Recommendation & Profile Service, which directly implement the assigned system's requirements.

UML DIAGRAM

